Armand Nicolicioiu

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Education

Oct. 2024 – **Ph.D. in Artificial Intelligence**
ETH Zürich, *Switzerland*
- Current research on repurposing generative models.
- Photogrammetry and Remote Sensing group.

Supervisors: Konrad Schindler and Răzvan Pașcanu (Mila & DeepMind)

2020 – 2022 **M.Sc. in Artificial Intelligence**
University of Bucharest, *Romania*
- Thesis: Research in out-of-distribution generalization by diversifying Attention heads in Vision Transformers.

Supervisor: Prof. Bogdan Alexe

2015 – 2020 **B.Sc. in Computer Science and Engineering**
Politehnica University of Bucharest, *Romania*
- Thesis: Research in meta-learning and few-shot classification using Graph Neural Networks and Attention mechanisms.

Supervisor: Prof. Marius Leordeanu

Work Experience

03/2024 – 08/2024 **Research Intern**
Institute of Science and Technology Austria (ISTA)
- Developed a personalized writing assistant running entirely on-device using **compressed LLMs**. Supervisor: Prof. Dan Alistarh.

10/2023 – 01/2024 **Research Intern**
ETH Zürich, *Switzerland*
- Research on **semi-supervised learning** in the Statistical Machine Learning group led by Prof. Fanny Yang.

05/2023 – 09/2023 **Research Intern**
Idiap Research Institute, *Martigny, Switzerland*
- Research on the **robustness** of **diversified** model ensembles and on the **generalization** properties of neural networks. Supervisor: Dr. Damien Teney.

11/2022 – 03/2023 **Software Development Engineering Intern**
Amazon, *Tübingen, Germany*
- Improved an evaluation framework for assaying **out-of-distribution generalization** in the Causal Representation Learning team.

06/2022 – 09/2022 **Software Engineering Intern**
Google, *Zürich, Switzerland*
- Used **large language models** to improve entity linking in YouTube videos.
- Generated **realistic synthetic data** to complement existing training sets.

Work Experience (continued)

10/2020 – 11/2021
Machine Learning Researcher
***dotLumen**, Cluj-Napoca, Romania*
- Prototyped a headset to **help blind persons** be more independent.
- Multi-modal **environment understanding** using Computer Vision.

07/2019 – 09/2019
Software Engineering Intern
***Microsoft**, Redmond, WA, USA*
- Designed and built a fast **generic trending system** for **large scale data**.
- **Microservices pipeline** for ingestion, processing, storing and retrieval.

07/2018 – 09/2018
Software Engineering Intern
***Microsoft**, Redmond, WA, USA*
- Improved Azure's **real-time telemetry** monitoring system by generating **recommendations** that lead to a **faster resolution of alerts**.

01/2017 – 11/2017
Machine Learning and Computer Vision Junior Researcher
***Arnia Software**, Bucharest, Romania*
- Developed algorithms for **improving the HDR camera mode** of phones.

07/2016 – 10/2016
Machine Learning Engineer Intern
***Sparktech Software**, Bucharest, Romania*
- Built a **recommender system** for a coupon site from scratch.

Research Publications

1. **A. Nicolicioiu**, E. Iofinova, E. Kurtic, *et al.*, "Panza: Design and analysis of a fully-local personalized text writing assistant," *arXiv preprint arXiv:2407.10994*, 2024.
2. D. Teney, **A. M. Nicolicioiu**, V. Hartmann, and E. Abbasnejad, "Neural redshift: Random networks are not random functions," *CVPR*, 2024 **(Oral Presentation)**.
3. **A. M. Nicolicioiu**, A. L. Nicolicioiu, B. Alexe, and D. Teney, "Learning diverse features in vision transformers for improved generalization," *ICML Workshop on Spurious Correlations, Invariance, and Stability (SCIS)*, 2023.
4. L. Scimeca, A. Rubinstein, **A. M. Nicolicioiu**, D. Teney, and Y. Bengio, "Leveraging diffusion disentangled representations to mitigate shortcuts in underspecified visual tasks," *NeurIPS Workshop on Diffusion Models*, 2023.
5. L. Scimeca, A. Rubinstein, D. Teney, S. J. Oh, **A. M. Nicolicioiu**, and Y. Bengio, "Shortcut bias mitigation via ensemble diversity using diffusion probabilistic models," 2024.
6. N. Miolane, M. Caorsi, U. Lupo, *et al.*, "Iclr 2021 challenge for computational geometry & topology: Design and results," *arXiv preprint arXiv:2108.09810*, 2021.

Awards and Achievements

2023 • **Poster Award & Scholarship** at Synapse AI Symposium by Bending Spoons in Milan for my research project on model diversity.

2022 • **Best Poster Award** at Romanian AI Days conference for presenting my Master's thesis on *Diversifying Vision Transformers for Out-of-Distribution Generalization*.

• **Travel Scholarship** at Synapse AI Symposium by Bending Spoons in Milan.

Awards and Achievements (continued)

2021 • **1st place** at the Computational Geometry & Topology Challenge at ICLR for our study on Topological Noise Invariant Features.

2020 • **Best Poster Award** at Eastern European Machine Learning Summer School (EEML) for presenting my Bachelor's thesis on *Few shot learning by features adaptation with Graph Neural Networks.*

2019 • **1st place** at Catalyst Coding Contest (algorithms contest).

2018 • **1st place** at Textract (ML Hackathon).

2017 • **1st place** ITFest (algorithms contest).

2016 • Qualified for the **ACM-ICPC** Southeastern European Regional Contest (SEERC).

• **Best Research Idea** at AI-MAS Summer School at UPB.

2013, 2015 • **Bronze Medal** at the Romanian National Olympiad in Informatics.

2012 • **Bronze Medal** at the Romanian National Olympiad in Physics.

2009 – 2015 • **Finalist** in National Olympiad in Mathematics (2009, 2012), Physics (2010-2012), Informatics (2012-2015).

Teaching Experience

Feb. 2025 – **Teaching Assistant**
***ETH Zürich**, Switzerland*
- Classes: **Multivariate Statistics and Machine Learning** and **Image-based Mapping**.

2016 – 2021 **Undergraduate Teaching Assistant**
***Politehnica University of Bucharest**, Romania*
- Classes: **Data Structures** and **Computer Programming**.